- Easy page manipulation

Minimum Viable Product

Version 1.0 only needs:

☑Read front PDF
☑Read back PDF
☑Reverse back pages
☑Interleave pages
☑Handle odd page counts
☑Write merged PDF

Everything else can come later.

For this project, I'd actually start with less than 100 lines of Python and get the merge engine working first. Once that's proven with your tax documents, we can build the validation features and GUI on top of it.

Phase 4 - Command Line Interface

```
python duplex_merge.py \
   front.pdf \
   back.pdf \
   output.pdf
```

Phase 5 – Logging

Front pages : 37
Back pages : 36

Back pages reversed.

Merging...

Output:
tax_complete.pdf

Total Pages:
73

Success

Dry Run

--validate

Checks:

- PDF readable
- page counts
- merge feasibility

without creating output.

Preserve Metadata

Copy:

Processing

Step 1

Read all pages from

Step 2

Reverse back pages

Example:

6 4 2

becomes

2 4 6

Step 3

front[0] back[0] front[1] back[1] front[2] back[2]

Step 4

Generate
merged.pdf

Phase 2 - Odd Page Handling

Sometimes documents contain an odd number of pages.

Example

1 2 3 4 5

Front Scan:
1 3 5